

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q18-0926

Euclid 138 kV
April 22, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0926. The Interconnection Customer proposes a 78.1 MW solar photovoltaic facility interconnecting at the Euclid 138 kV substation in the New Jersey area. The request was received on January 22, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 361 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$3,609,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Camp Evers – Linden Cogen Plant 138.0 kV
Contingency: Pipe – Linden Cogen Plant 138.0 kV
Loading: 150.9% (Rating: 248 MVA)
DFAX: 8.7%
Pre-project loading: 148.1%
2. Facility: Linden Cogen Plant – Pipe Substation 138.0 kV
Contingency: Camp Eves – Linden Cogen Plant 138.0 kV
Loading: 130.9% (Rating: 261 MVA)
DFAX: 8.5%
Pre-project loading: 128.3%
3. Facility: Crawfordsville Power Plant – Pine Tree Acres 138.0 kV
Contingency: Dam Hts – Babine Lake 138.0 kV
Loading: 118.7% (Rating: 502 MVA)
DFAX: 11.5%
Pre-project loading: 116.9%
4. Facility: Lafayette Road Station – Secord 138.0 kV Line
Contingency: Crawfordsville Power Plant 345/138kV Transformer Bank
Loading: 110.9% (Rating: 286 MVA)
DFAX: 21.1%
Pre-project loading: 105.1%
5. Facility: Lafayette Road Station – Metuchen 138.0 kV
Contingency: Crawfordsville Ower Plant 345-138 kV Auto
Loading: 108.0% (Rating: 327 MVA)
DFAX: 21.1%
Pre-project loading: 102.9%

6. Facility: Allston – Metuchen 138.0 kV Line
 Contingency: Crawfordsville Power Plant 345/138 kV Autotransformer
 Loading: 105.8% (Rating: 366 MVA)
 DFAX: 21.1%
 Pre-project loading: 101.3%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Camp Evers – Linden Cogen Plant 138.0 kV L	Rebuild line with higher-capacity conductor	\$705,000	41% (\$291,000)
Linden Cogen Plant – Pipe 138.0 kV	Rebuild line with higher-capacity conductor	\$663,000	100% (\$663,000)
Crawfordsville Power Plant – Pine Tree Acres 138.0 kV	Rebuild line with higher-capacity conductor	\$931,000	38% (\$357,000)
Lafayette Road Station – Secord 138.0 kV Line	Rebuild line with higher-capacity conductor	\$3,122,000	54% (\$1,690,000)
Lafayette Road Station – Metuchen 138.0 kV	Reconductor line	\$321,000	54% (\$174,000)
Metuchen – Allston 138.0 kV L	Reconductor line	\$802,000	54% (\$434,000)
	Total allocated to project		\$3,609,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.